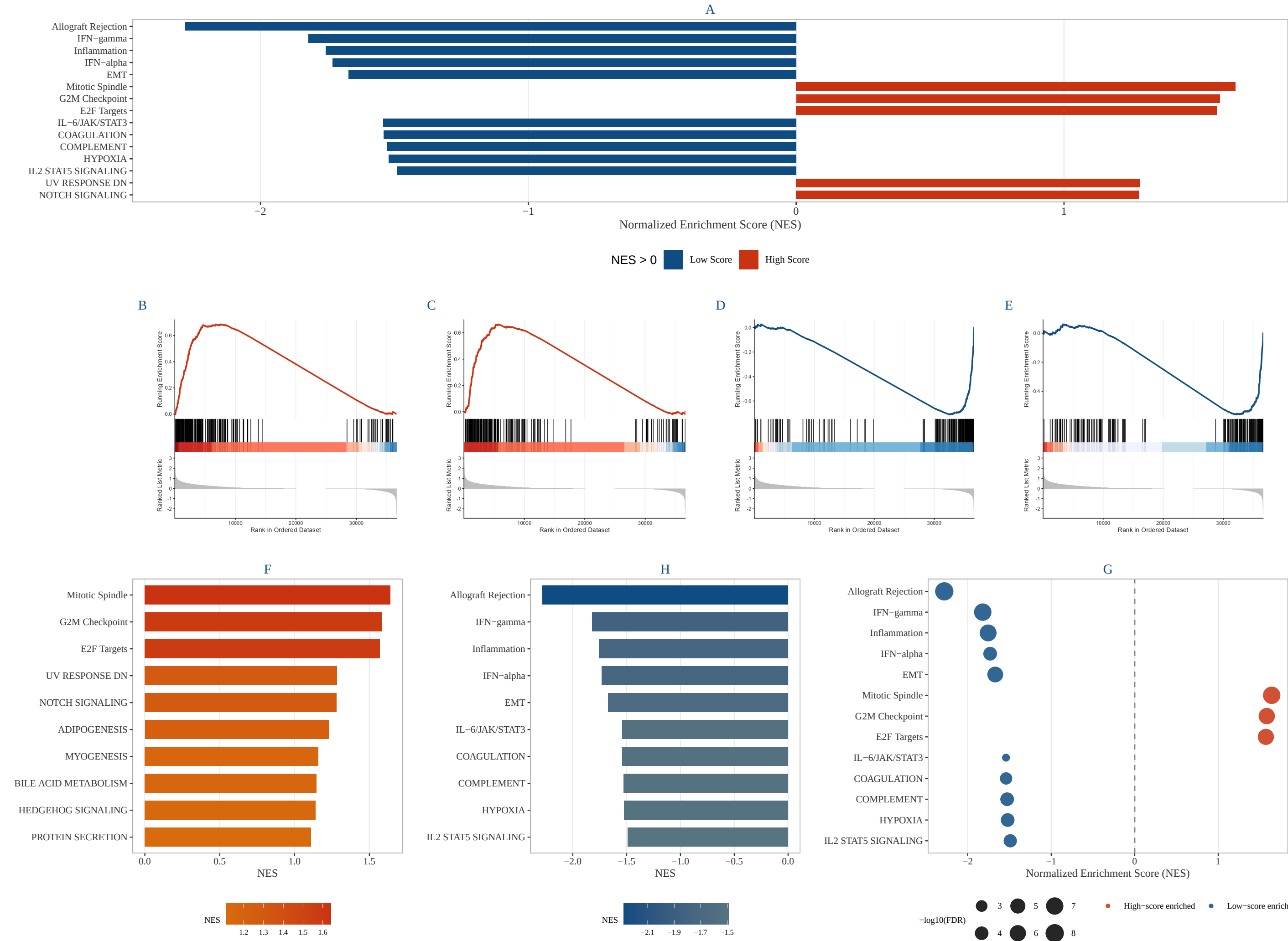


Figure 4: GSEA Dual Phenotype

A) NES bar plot (Top 15) | B-E) Representative GSEA curves | F) Up-regulated pathways | H) Down-regulated pathways | G) Forest plot of significant pathways (FDR < 0.05)



Note: All plots based on real GSEA results from clusterProfiler.
Red: enriched in high-score group; Blue: enriched in low-score group.
Forest plot point size represents $-\log_{10}(\text{FDR})$.